

PYTHON ASSIGNMENT – 4

MINI PROJECT

Course: Python for Data Science

Academic Year: 2025 – 2026 [Even Semester]

Title: Election Result Data Visualization System

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INTRODUCTION

Election Result Data Visualization System is an advanced data analysis project developed using Python libraries such as NumPy, Pandas, and Matplotlib. Elections generate huge amounts of data including voter statistics, constituency-wise results, party performance, candidate information, and voting percentages. Managing and analyzing this information manually is difficult and time-consuming. This project focuses on collecting, processing, and visualizing election-related data effectively. The system helps users understand election trends through charts, tables, and graphical analysis. It provides meaningful insights about winning parties, vote percentages, regional performance, and voter turnout. The project demonstrates practical implementation of data science concepts such as data cleaning, aggregation, grouping, merging, filtering, and visualization. The system improves understanding of real-world data analysis applications and helps students gain practical exposure to Python programming and analytical techniques.

OBJECTIVE

- 1. To analyze election result datasets efficiently.
- 2. To perform data cleaning and preprocessing operations.
- 3. To visualize election trends using graphs and charts.
- 4. To identify winning parties and candidates.
- 5. To calculate vote percentages and voter turnout.
- 6. To generate constituency-wise reports.
- 7. To apply NumPy and Pandas concepts in real-world election analysis applications.

SOFTWARE REQUIREMENTS

Software	Version
Python	3.x
NumPy	Latest
Pandas	Latest
Matplotlib	Latest
Jupyter Notebook / VS Code	Any

PROBLEM STATEMENT

Managing election data manually is difficult because elections involve large amounts of constituency-wise and candidate-wise information. The objective of this project is to develop an Election Result Data Visualization System using NumPy and Pandas that can:

- Store election result records

- Analyze vote distribution
- Identify winning candidates and parties
- Calculate voter turnout percentages
- Generate constituency-wise reports
- Display election trends through visualizations

DATASET DESCRIPTION

Column Name	Description
Candidate_ID	Unique candidate number
Candidate_Name	Name of the candidate
Party	Political party name
Constituency	Election constituency
Votes_Received	Total votes secured
Voter_Turnout	Percentage of voter turnout
Election_Date	Date of election

SYSTEM DESIGN

Modules

1. Data Collection

Election data is collected and stored in CSV format.

2. Data Cleaning

Missing values and duplicate records are removed for better accuracy.

3. Data Processing

The collected data is processed using NumPy and Pandas libraries.

4. Analysis Module

The system performs various analyses including party-wise analysis, constituency analysis, and voter turnout analysis.

5. Visualization Module

Graphs and charts are generated using Matplotlib to display election trends clearly.

ALGORITHM

Step-by-Step Algorithm

1. Import NumPy and Pandas libraries.
2. Read election dataset using Pandas.
3. Display first few rows of the dataset.
4. Check for missing values.
5. Fill missing values using fillna().
6. Remove duplicate records.
7. Perform grouping based on political parties.
8. Calculate vote percentages.
9. Generate pivot tables.
10. Plot charts using Matplotlib.
11. Display analysis results.

PYTHON PROGRAM

```
# Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Create Sample Dataset
election_data = {
    'Candidate_ID':[101,102,103,104,105],
    'Candidate_Name':['Arun','Bala','Charan','Divya','Eswar'],
    'Party':['ABC','XYZ','ABC','LMN','XYZ'],
    'Constituency':['North','South','East','West','Central'],
    'Votes_Received':[45000,52000,47000,39000,50000],
    'Voter_Turnout':[72,68,75,70,74]
}

# Create DataFrame
df = pd.DataFrame(election_data)

# Display Dataset
print(df)
```

```
# Group By Party
party_analysis = df.groupby('Party')['Votes_Received'].sum()

# Plot Bar Graph
party_analysis.plot(kind='bar')
plt.title('Party Wise Vote Analysis')
plt.xlabel('Party')
plt.ylabel('Votes')
plt.show()
```

OUTPUT

The Election Result Data Visualization System successfully analyzed election datasets and generated graphical representations of voting patterns. The system displayed party-wise vote distribution, constituency performance, and voter turnout analysis through bar charts and line graphs.

RESULT

The Election Result Data Visualization System was implemented successfully using NumPy and Pandas. The system effectively:

- Managed election records
- Analyzed party-wise performance
- Calculated vote percentages
- Generated constituency-wise reports
- Displayed graphical visualizations

Thus, the project demonstrates the practical implementation of data analysis concepts in election management systems.

CONCLUSION

This project helped in understanding real-world election data analysis using Python libraries. Students learned about DataFrames, grouping operations, aggregation techniques, pivot tables, and visualization concepts. The project provides practical exposure to NumPy and Pandas concepts included in the syllabus and improves analytical thinking skills in handling election datasets.

FUTURE ENHANCEMENT

1. Real-time election result tracking
2. Web application integration
3. AI-based election prediction system
4. Mobile application support
5. Interactive dashboard creation
6. Database connectivity
7. Advanced data visualization techniques